Design LCC Rpi Pico Shield

Raspberry Pi Pico

March 20, 2026

	0	1	2
ADC			
SPI			
I2C			
UART			

					2.3 Stepper board	2.4 Stepper board	2.5 Node board	2.6 Node board	2.7 Node board	2.7 Stepper board	2.8 Node board
1	gp0	SPI0 RX	I2C0 SDA	UART0 TX	LED 1 / Blue Button	Bridge sensor		header pin	I/0-1:Pin1	TT-Pin3 - Bridge sensor	ACAN_RX_PIN
2	gp1	SPI0 CSN	I2C0 SCL	UART0 RX	LED 2 / Gold Button	Home sensor		header pin	I/0-1:Pin2	TT-Pin2 - Home sensor	ACAN_CS_PIN
3	GND										
4	gp2	SPI0 SCK	I2C1 SDA		Touch Reset	Stepper Direction		Npixel 1	I/0-1:Pin3	Stepper Direction	ACAN_CSK_PIN
5	gp3	SPI0 TX	I2C1 SCL		Touch Interrupt	Touch Interrupt		Npixel 2	I/0-1:Pin4	TT-Pin4 - NeoPixel data line	ACAN_TX_PIN
6	gp4	SPI0 RX	I2C0 SDA	UART1 TX	Touch SDA	Touch SDA		I2C0 SDA	I/0-1:Pin7 / I2C0 SDA	header pin	ACAN_INT_PIN
7	gp5	SPI0 CSn	I2C0 SCL	UART1 RX	Touch SCL	Touch SCL		I2C0 SCL	I/0-1:Pin8 / I2C0 SCL	header pin	header pin
8	GND										
9	gp6	SPI0 SCK	I2C1 SDA		Display Data 0	Display Data 0		Npixel 3	I/0-1:Pin9	Stepper Enable	I2C1 SDA
10	gp7	SPI0 TX	I2C1 SCL		Display Data 1	Display Data 1		Npixel 4	I/0-1:Pin10	Stepper Step	I2C1 SCL
11	gp8	SPI1 RX	I2C0 SDA	UART1 TX	Display Data 2	Display Data 2		I/O - Pin10	I/0-2:Pin1	I/0-2:Pin1 : SPI1 SDI	I/0-1:Pin1
12	gp9	SPI1 CSn	I2C0 SCL	UART1 RX	Display Data 3	Display Data 3		I/O - Pin9	I/0-2:Pin2	I/0-2:Pin2 : SPI1 CS	I/0-1:Pin2
13	GND										
14	gp10	SPI1 SCK	I2C1 SDA		Display Data 4	Display Data 4		I/O - Pin8	I/0-2:Pin3	I/0-2:Pin3 : SPI1 CLK	I/0-1:Pin3
15	gp11	SPI1 TX	I2C1 SCL		Display Data 5	Display Data 5		I/O - Pin7	I/0-2:Pin4	I/0-2:Pin4 : SPI1 SDO	I/0-1:Pin4
16	gp12	SPI1 RX	I2C0 SDA	UART0 TX	Display Data 6	Display Data 6		I/O - Pin4	I/0-2:Pin7	I/0-2:Pin7 : open	I/0-1:Pin7
17	gp13	SPI1 CSn	I2C0 SCL	UART0 RX	Display Data 7	Display Data 7		I/O - Pin3	I/0-2:Pin8	I/0-2:Pin8 : Touch INT	I/0-1:Pin8
18	GND										
19	gp14	SPI1 SCK	I2C1 SDA		Stepper Enable	Stepper Enable		I/O - Pin2	I/0-2:Pin9	I/0-2:Pin9 : Touch I2C0 SDA	I/0-1:Pin9
20	gp15	SPI1 TX	I2C1 SCL		Stepper Step	Stepper Step		I/O - Pin1	I/0-2:Pin10	I/0-2:Pin10 : Touch I2C0 SCL	I/0-1:Pin10
21	gp16	SPI1 RX	I2C0 SDA	UART0 TX	TX	ACAN_RX_PIN		ACAN_RX_PIN	ACAN_RX_PIN	ACAN_RX_PIN	I2C0 SDA
22	gp17	SPI1 CSn	I2C0 SCL	UART0 RX	RX	ACAN_CS_PIN		ACAN_CS_PIN	ACAN_CS_PIN	ACAN_CS_PIN	I2C0 SCL
23	GND										
24	gp18	SPI1 SCK	I2C1 SDA		Bridge sensor	ACAN_CSK_PIN		ACAN_CSK_PIN	ACAN_CSK_PIN	ACAN_CSK_PIN	I/0-2:Pin1
25	gp19	SPI1 TX	I2C1 SCL		Home sensor	ACAN_TX_PIN		ACAN_TX_PIN	ACAN_TX_PIN	ACAN_TX_PIN	I/0-2:Pin2
26	gp20		I2C0 SDA		Stepper Direction	ACAN_INT_PIN		ACAN_INT_PIN	ACAN_INT_PIN	ACAN_INT_PIN	I/0-2:Pin3
27	gp21		I2C0 SCL		NeoPixel data line	NeoPixel data line		Blue Button	Blue Button	Blue Button	I/0-2:Pin4 & Blue Button
28	GND										
29	gp22				Display WR	Display WR		Gold Button	Gold Button	Gold Button	I/0-2:Pin7 & Gold Button
30	RUN					Reset Button		Reset Button / header	Reset Button / header	Reset Button / header	Reset Button / header
31	gp26	ADC0	I2C1 SDA		I2C1 SDA	I2C SDA / Blue Button		I2C1 SDA	I2C1 SDA	I2C1 SDA	I/0-2:Pin8
32	gp27	ADC1	I2C1 SCL		I2C1 SCL	I2C SCL / Gold Button		I2C1 SCL	I2C1 SCL	I2C1 SCL	I/0-2:Pin9
33	GND	A GND						header pin	header pin	header pin	I/0-2:Pin7 alternative
34	gp28	ADC2			Display D/C	Display D/C		header pin	header pin	header pin	I/0-2:Pin10
35		ADC_VREF						header pin	header pin	header pin	I/0-2:Pin6 alternative
36	3v3_OUT										
37	3V3_EN										
38	GND										
39	VSYS										
40	VBUS										

GND		I/0-1:Pin5	I/0-1:Pin5
Vselect1		I/0-1:Pin6	I/0-1:Pin6
GND		I/0-2:Pin5	I/0-2:Pin5
Vselect2		I/0-2:Pin6	I/0-2:Pin6